Deakin University

Research Project/Research Project (Advanced)

OnTrack Submission

Mid-term Research Presentation

Submitted By:
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2026/04/14 13:05

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Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO5 Application of Skills and Knowledge

April 14, 2026

Detection and Simulation of GPS Spoofing in Mobile Robots

Mid-term Research Presentation — Task 4.3D

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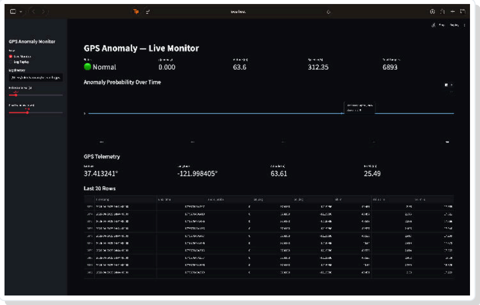
Deakin University   Supervisor: Dr. Ananda Maiti

April 2026

Primary Objective

Develop a real-time GPS spoofing **detection system** using **ML-based sensor cross-validation**.

- Live telemetry monitoring.
- GPS $\times$ IMU validation.
- Dynamic **Trust Index** (0–1).

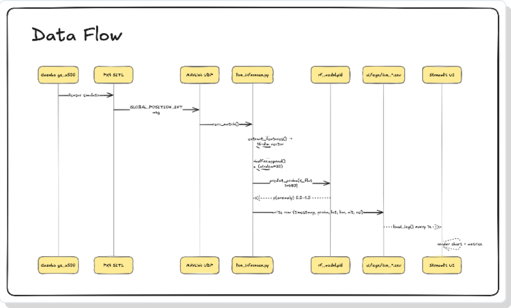


Live ML Inference Dashboard

Literature Review & Research Gaps

Identified Gaps:

- Gap 1: Real-time implementation on embedded hardware.
- Gap 2: Continuous reliability scoring for navigation.
- Gap 3: Breaking Point analysis under variable spoofing.



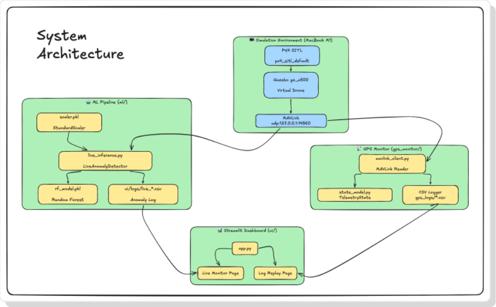
System Information Architecture

Research Questions

RQ1 What are the primary vulnerabilities of GPS in mobile robots?

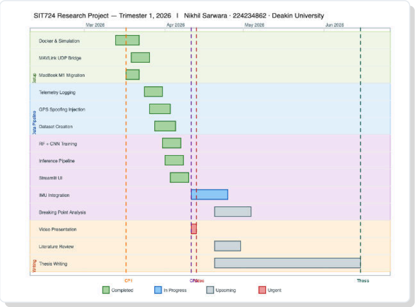
RQ2 How can sensor fusion improve real-time ML detection?

RQ3 How can a **Trust Index** quantify reliability?



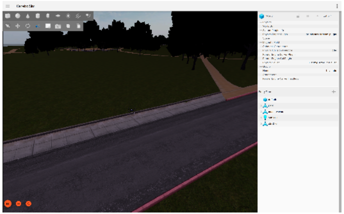
Research Plan (Inc. SIT723)

Phase	Key Activities	Status
SIT723	Literature Review & Env Setup	Complete
Mar 2026	MAVLink UDP Bridge & Inspector Tool	Complete
Apr 2026	ML Model Training (RF/CNN) & UI	Complete
May 2026	IMU Integration & Breaking Point	In Progress



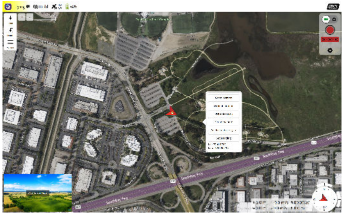
Current Progress: Live Simulation Dashboard

Simulation Env



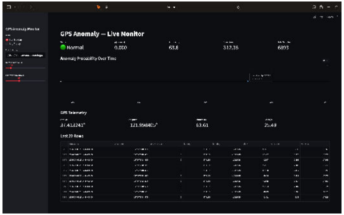
PX4 SITL in Gazebo

Robot Control



QGroundControl

ML Inference



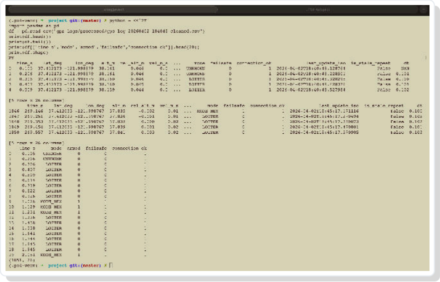
Live GPS Trust Index

Key Result

Bypassed Docker constraints with a custom MAVLink bridge for high-speed ML inference.

Challenges & Mitigations

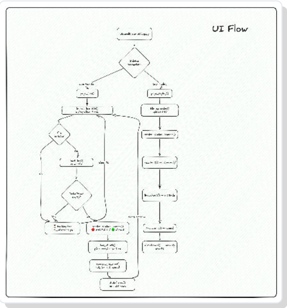
- **PX4 Change:** Scripts broke due to jMAVSim removal.
- *Mitigation:* Designed **MAVLink UDP Bridge**.
- **Telemetry Noise:** Raw data labeling difficulty.
- *Mitigation:* Automated JSON-to-CSV pre-processing.



Cleaned Dataset Analysis

Next Steps

1. Integrate **IMU data** into fusion logic.
2. Quantify the **Breaking Point** of detection.
3. Finalize Task 2.1 (Methodology).



Thank You

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